Ashwin Chembu

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PROFILE

Full-stack product engineer connecting React and TypeScript interfaces with Python and Node.js services, authentication, and cloud data. Builds complete workflows with role permissions and layered validation.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Frontend: React, Next.js, Tailwind CSS, Capacitor, responsive interfaces

Backend / APIs: Node.js, Express, FastAPI, REST APIs, OAuth, WebSockets

Data / Cloud: MongoDB, Firestore, Firebase Authentication, AWS S3

Testing: Vitest, React Testing Library, Firebase emulators, Playwright

EXPERIENCE

Business Technology Solutions Associate | ZS Associates **South San Francisco, CA | Jun 2024-Present**

- Built **React**/FastAPI RAG assistants with AWS document ingestion and Redis services for enterprise knowledge search.
- Built Python/Flask endpoints connecting AWS S3 and Smartsheet APIs to automate document storage and cloud handoffs.
- Translated stakeholder needs into testable rules, API integrations, source-to-target mappings, and delivery documents.
- Contributed FastAPI/Uvicorn model workflows and pytest checks to validate enterprise RAG services and AI behavior.

Founder and Lead Developer | Tallyrus **Jun 2023-Sep 2025**

- Designed **React** rubric interfaces and feedback loops so educators could review essays with consistent scoring logic.
- Integrated **React**/**TypeScript** and Express flows with Google OAuth, MongoDB, sessions, and protected product routes.
- Built Flask services with MongoDB persistence and S3 storage to manage documents and support essay-feedback flows.

Data Analyst / Web Development Lead | Tech4Good at UCSC **Mar 2023-Jun 2025**

- Built **TypeScript**/Figma API chat, polling, and export workflows in a collaborator-owned repository for classroom collaboration.
- Implemented 20+ attributed Node/Express routes with MongoDB persistence and logging for chat, text entry, and polls.
- Separated chat, text, and poll state in three MongoDB databases, using change streams to synchronize edits and votes.

PROJECTS

Sylk | Multi-Tenant Supply-Chain Platform **Private organization product | 2025-Present**

- Built **React**/**TypeScript** shipment flows with **Firebase** for onboarding, line items, fulfillment, files, and real-time chat.
- Enforced tenant isolation with Firestore/Storage rules, separating retailer finances from shared manufacturing flows.
- Tested tenant access and supply-chain journeys with Vitest, Testing Library, **Firebase** emulators, and **Playwright.**
- Packaged a Capacitor Android app with native-safe routing, **Firebase** Authentication, and phone-specific navigation.

Lume | AI Meeting Intelligence **Dec 2025-Jan 2026**

- Built **React** and Node/Express flows with MongoDB/S3 to capture, store, search, and query meetings and project files.
- Designed RAG with vector retrieval, intent routing, evidence extraction, and generator/critic passes for cited answers.
- Connected meeting capture, document search, and reusable context through responsive **React** and WebSocket updates.

3D Commerce Experience | Interactive Storefront **Public UI prototype**

- Built reusable **React**/**TypeScript** storefront components for discovery, search, cart, checkout, account, and content pages.
- Built a Three.js configurator with GLB/Draco assets and independent garment-color, material, and placement controls.
- Built navigation, product cards, modals, and consent flows with Vite/Tailwind; verified a successful production build.

EDUCATION

University of California, Davis | **B.S. Computer Science** | **GPA: 3.7** Graduated Jun 2025

CERTIFICATIONS

AWS Certified Cloud Practitioner | Google Data Analytics Certificate

LINKS

Portfolio: https://ashchembu.com | LinkedIn: linkedin.com/in/ashwin-chembu-a5292b205 | GitHub: github.com/ashwinchembu